

NetSI PhD Bootcamp 2026 · Thursday, Sep 3 · 2pm–4pm

Agentic Computing

Coding agents in your PhD workflow

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How this session works

- ~50 min guided speed-run: setup, the basics, and how to drive an agent well
- ~55 min **Hackathon** pick one of three projects to build
- ~15 min share-outs

Setup (do this now)

Hands-on Session

1. Go to claude.northeastern.edu, click **First time? Start here**, accept the guidelines
2. Log in with **SSO** (your NU email + credentials)
3. Install **Claude Code**: code.claude.com/docs/quickstart (terminal + desktop)

Generous NU usage limits. Prefer something else? OpenAI Codex or opencode (open source) work too.

Open both: the GUI and the terminal

Hands-on Session

- **Claude Desktop** (the GUI): chat, drag in files and images, quick questions
- **Claude Code** (the terminal): run `c̣laude` inside a project, it reads and edits your files and runs commands
- Same account, same models. Use the GUI to think, the terminal to build

Basics

What is an agent?

An LLM that can read files, run commands, and edit code, in a loop

- You give a **goal**; it plans, acts, checks its own work, and repeats
- It works on your **whole project**, in your terminal, not just one snippet
- You stay the reviewer: it drafts, **you** decide what to keep

Think pair-programmer that can also run the tests and read the errors.

CLAUDE.md: your project's system prompt

Standing instructions the agent reads **every** session

- `/init` drafts one from your codebase. Keep it short: stack, how to run things, do's and don'ts
- The more it knows about your project, the better it performs
- Curious what strong system prompts look like? Read Anthropic's published ones: platform.claude.com/docs

CLAUDE.md

- Python 3.11, networkx + matplotlib. Submit with ``sbatch``, never compute on the login node.
- Small diffs. Show me a plan before editing more than one file.

Which model?

Pick the brain for the job (switch with `/model`)

- **Opus** (4.8, 5): your **default**. Planning, implementation, hard reasoning, gnarly bugs
- **Sonnet**: faster and cheaper, for basic or mechanical tasks
- **Haiku**: cheapest, for trivial tasks

Rule of thumb: **use Opus for almost everything**, drop to Sonnet only for very basic things.

Track everything with git

Commit often so you can always go back

- **Commit frequently**, and **push** to a remote (GitHub) as an off-machine backup
- `git diff` to review what the agent changed; `git restore <file>` to undo just one file
- **Browse the history visually** in VS Code's **Timeline** view (Explorer sidebar), no `git log` needed
- Let the agent run git for you, but understand each step. That is what makes experimenting safe

Text is supreme: work in markdown

Notes, ideas, plans, and docs, all as plain `.md`

- Store your thinking where the agent can read it: notes, half-baked ideas, plans, design docs
- Text is **versionable**, **diffable**, and **agent-native**. A plan in markdown beats a plan in your head
- *L^AT_EX* source is text too 😊

Driving the agent

Start a task: seed context, then brainstorm

- `/init <idea>` to give the agent a starting point
- **Brainstorm here.** Throw new, not-fully-formed thoughts at it and talk them through
- This early back-and-forth builds the context that makes everything after it better

Plan before you build: `/plan`

Planning is the highest-leverage step

- It asks **you** questions. **Iterate over the plan several times** before any code
- Later: skills like `/grill-me` push you with harder questions
- **You** make the decisions, not the agent, especially:
 - architecture and design
 - anything that is messy to undo once you start

Stay in control

- Press `esc` to stop it mid-run: **stop and steer**. Talk to it the moment you see a mistake
- It sounds **confident even when it is wrong**. You own every command it runs, read before you approve
- **Never** put secrets (keys, passwords, private data) into a prompt
- **Do NOT use** `--dangerously-skip-permissions` yet. It acts without asking and can wipe a week's work. Only once you are confident and the task is safe

Skills: reusable instructions

Packaged text for tasks you repeat

- A skill teaches a workflow once, then you reuse it: `review`, learn a package, a house style
- Have the agent **mine your history** for skills worth extracting
- Skills enrich context, and more context means better performance
- Examples: `presentations`, `grill`, `review`. Build your own

Subagents: clones with a fresh start

A helper agent that starts from only what the parent tells it

- Great for a self-contained side task (a review, a focused search)
- They burn **lots of tokens**, and it grows fast the more you spawn. Use them judiciously
- Match the model to the job:
 - rename files ▶ Sonnet/Haiku
 - research overview ▶ Opus

Using it for research

Learning

Use a strong model here (Opus-level)

- **Explain it at my level:** "like I'm 5 / an undergrad / a postgrad / an expert"
- **Deep research** (Claude, ChatGPT, Gemini): let it scrape widely and hand you an overview
- [NotebookLM](#): drop in your papers, get grounded answers and an audio overview
- **Starting a topic?** Map the area, name the seminal papers, get a sensible reading order
- Prompt ideas: [Wharton GenAI prompt library](#)

Prototyping

The cost of **trying** an idea is now near zero

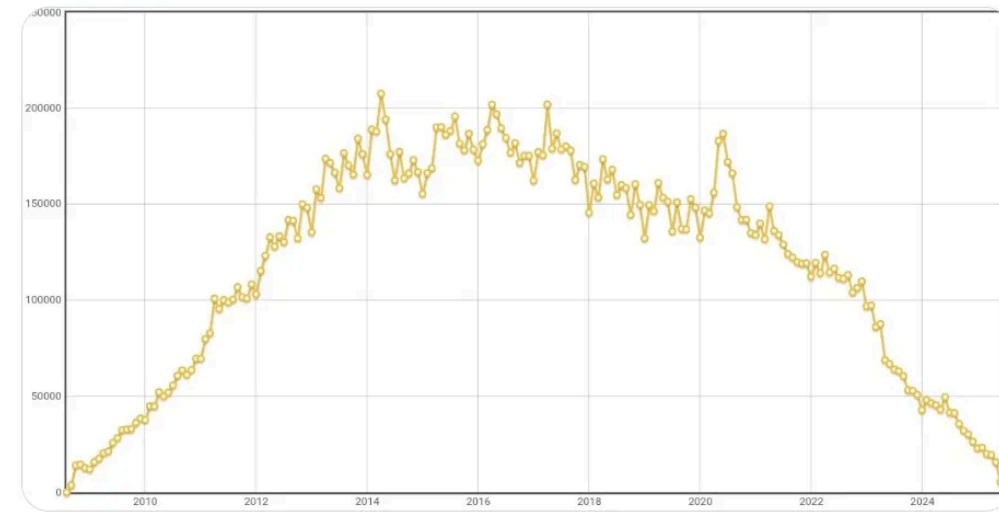
- Spin up a **quick-and-dirty prototype** of a research or project idea. If it works, go back, understand **why**, then clean and extend it
- **Escape dependency hell** on the cluster (story time: ask Minami)
- **Debugging?** Use it like Stack Overflow or Google: paste the error, get unstuck



Marc Gravell
@marcgravell



StackOverflow questions over time, source SEDE; sadface, lunch has been eaten



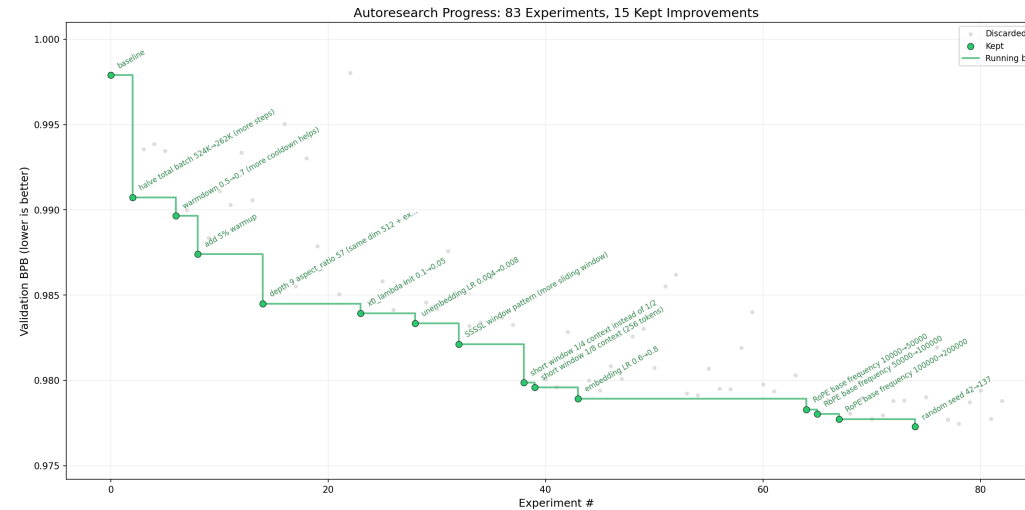
9:51 AM · May 15, 2025 · 65K Views

Understanding and quality

- **Review and grill:** ask it to review the work, then **grill it with questions, iteratively**
- **Testing:** it writes unit tests fine, but does not know your domain well enough to test what **matters**. Give it more context (talk to it, add papers, edit `CLAUDE.md`)
- **Visual testing:** for plots and UIs, tell it to **look at the image** and fix overlaps, bad colors, obvious glitches (it can see images)
- **Refactoring:** better names, readable code, split concerns (which also makes it testable)

The frontier: auto research

Set a goal and guardrails, then let it loop overnight



Karpathy's [autoresearch](#): an agent edits `train.py`, runs ~100 five-minute experiments overnight, keeping the ones that lower validation loss (`val_bpb`).

Reality check: it drifts, over-claims, and burns compute. It's an assistant you need to **supervise**; **you** are the scientist. Verify everything.

OpenAI

OpenAI – Hugging Face Incident

Technical Report

Part 2: Hackathon

Three paths. Pick yours.

Choose your project

Track	Difficulty	You'll build
Dashboard website	BEGINNER	a static page that visualizes a network
Extend Session 1	INTERMEDIATE	more from your morning's cluster track
Recreate a paper	INTERMEDIATE	a result from a network science paper

Prompts, starter files, checklists: [hackathon/agentic/](https://hackathon.agentic/)

Track 1: Dashboard website

BEGINNER

A static site that visualizes a network, built by the agent

- Loads a graph, draws it (nodes sized by degree), shows stats + a degree histogram
- Plain HTML + JS (vis-network or D3), no build step. Starter `network.json` included
- Great on-ramp: the focus is the **workflow**, not hand-writing web code

▶ hackathon/agentik/track-1-dashboard/

Track 2: Extend your Session 1 work

INTERMEDIATE

Take this morning's cluster track further, with the agent

- Track A ▶ 2-D parameter sweep + heatmap
 - Track B ▶ sweep GNN architectures
 - Track C ▶ add an igraph comparison, or push the size ladder higher
 - Practice: a cluster-aware `CLAUDE.md`, `sbatch --test-only`, verify the real job output
- ▶ [hackathon/agentic/track-2-extend-session1/](https://hackathon.agentic.track-2-extend-session1/)

Track 3: Recreate a paper

INTERMEDIATE

Build this hierarchical network, then reproduce its result plots

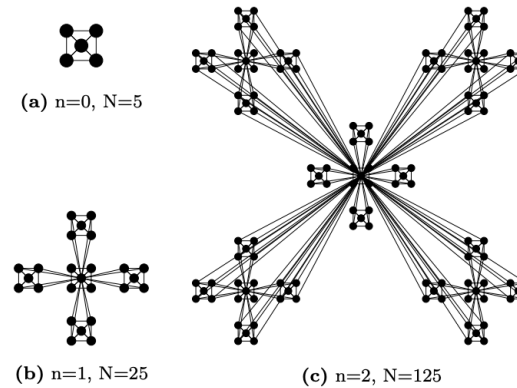
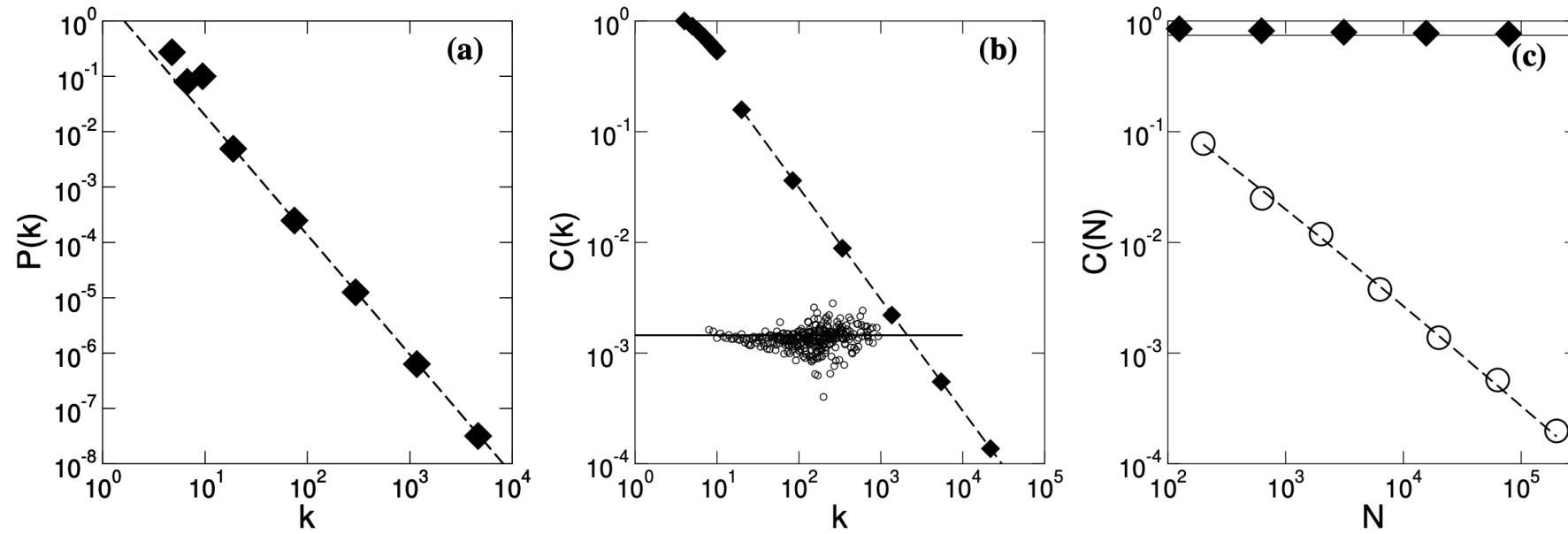


FIG. 1: The iterative construction leading to a hierarchical network. Starting from a fully connected cluster of five nodes shown in (a) (note that the diagonal nodes are also connected – links not visible), we create four identical replicas, connecting the peripheral nodes of each cluster to the central node of the original cluster, obtaining a network of $N = 25$ nodes (b). In the next step we create four replicas of the obtained cluster, and connect the peripheral nodes again, as shown in (c), to the central node of the original module, obtaining a $N = 125$ node network. This process can be continued indefinitely.

► hackathon/agentik/track-3-recreate-paper/ · arXiv [cond-mat/0206130](https://arxiv.org/abs/cond-mat/0206130)

The plots to reproduce



Your targets: **(a)** scale-free $P(k)$ · **(b)** $C(k) \sim 1/k$, the hierarchy fingerprint · **(c)** clustering stays constant as N grows

Share-outs

60 seconds each

- **What I tried** (the goal)
- **What worked**
- **What broke**, and how you dealt with it
- **What I'd try next**

Takeaways

- Agents are a **force-multiplier, not autopilot**. You plan and verify
- Most of the value: a tiny `CLAUDE.md` + small, verifiable steps
- Best combo: **cluster + agent**. You did both today

Go use it on your actual research tomorrow.